

The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap for a Concrete Transfer Operator

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DRAFT — not for distribution; live slots present

Abstract

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$, we machine-check in Lean 4 that the Osterwalder–Schrader (site-form) reconstruction of the transfer operator carries *one* mass: there is a single $m > 0$ such that for *every* spatial extent L the reconstructed (site-form) transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised kernel whose projected norm the Dobrushin corollary bounds by e^{-m} , and the reconstructed two-point functions are identified n -step against the Gibbs measure with the volume-uniform rate **[SLOT: B3 clause shape as proved]**. All quantifiers live in the Lean statements; the prose here copies the kernel’s cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data $\rightarrow$ physical Hilbert space $\rightarrow$ self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (GappedTransfer); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

1 Introduction and landscape

[SLOT: 2–3 paragraphs: the reconstruction idea; why volume-uniformity is the content (a gap at every fixed extent is classical and already in the programme’s paper 8; ONE m for all extents is what a thermodynamic limit can consume); the D-6 input.]

1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: docs/OS-R-PRIORITY-COLLATION.md in the repository).

*With machine-checked artefacts produced on the programme’s Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, anchor **[SLOT: anchor SHA of the green module]**.

Artefact	What is machine-checked	What is not
<code>mrdouglasny/reflection-positivity</code> (sorry-free ~2026-06)	$\text{Positivity} \rightarrow$ Hilbert space (completion/quotient) $\rightarrow$ self-adjoint contraction T ; trace formulas; susceptibility bounds from a gap	the gap itself: carried as the GappedTransfer hypothesis ; no spatial-volume-uniform statement; concrete instance is finite-lattice RP only
<code>OSforGFF</code> (arXiv:2603.15770)	all five GJ/OS axioms for the $d=4$ Gaussian free field, 0 sorries	reconstruction (listed as next milestone); any interacting model
<code>xiyin137/OSreconstruction</code>	Wightman-side GNS lane of the full $\text{OS} \leftrightarrow$ Wightman theorem	the theorem: 53 sorries + 12 axioms at last README snapshot
<code>mrdouglasny/lgt</code> (complete 2026-05-04)	Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on (n, d, β)	an operator gap; any reconstruction; the statement is per-volume on the measure side
<code>pphi2</code>	transfer-matrix spectral machinery for φ_2^4	rests on 27–29 project axioms
This paper	[SLOT: one-line summary of the green endpoints]	Wightman axioms; continuum limit; $SU(N)$; Clay

2 The two geometries and the forced operator

[SLOT: Cite the programme modules: `SpatialReflection` (paper 12), `SpatialOS` (paper 13), `SpatialReconstruction` (paper 14) — the site form weighs by $1/w$, the forced operator $T = D_w K$, self-adjoint at every β , PSD for $\beta \geq 0$. State the defining equation as the Lean theorem names. NO new mathematics in this section.]

3 The dressing unitary

All three statements are green on the plane (module `YangMills/OS/OSReconstructionUniform.lean`, sha256 5445429f . . . , five-pass ladder, `ELAB_EXIT 0`). Write $Qu := \sqrt{w} u$ for the boundary dressing of a real Euclidean vector (`qEmbed`); the site form weighs by $1/w$.

Theorem 3.1 (`A1, siteForm_qEmbed`). For $w > 0$ and real u, v : $\text{siteForm } w (Qu) (Qv) = \sum_{\sigma} u_{\sigma} v_{\sigma}$. Per site the cancellation is elementary: $\sqrt{w} u \cdot \sqrt{w} v \cdot (1/w) = uv$.

Theorem 3.2 (`A2, transferOp_qEmbed`). $T(Qu) = Q(\text{act}(\text{symWeighted } w \beta) u)$ pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel’s action.

Theorem 3.3 (`A3, transferOp_iterate_qEmbed`). $T^{[n]}(Qu) = Q((\text{act } S)^{[n]} u)$ — the shape the in-tree measure bridge `gibbsPathSum_eq_iterate` consumes.

4 The transported volume-uniform gap

Theorem 4.1 (`os_reconstruction_uniform_gap`). *In the window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$ there is $m > 0$ such that for every L there are $\lambda > 0$ and $\Omega > 0$ with the fixed-point equation $\sum_{\tau} \text{tiltKernel}(\text{sliceW } \gamma L) \beta \lambda \sigma \tau \Omega_{\tau} = \Omega_{\sigma}$, the projected bound $\|\text{projectedTransfer} \dots\| \leq e^{-m}$, and the A2 identity at $w := \text{sliceW } \gamma L$.*

The proof is an obtain on `dobrushin_ising_uniform_gap` followed by A2: *the same witnesses*, no new analytic content — which is precisely the point. The uniformity is real because it is D-6’s; the reconstruction contributes the identification, not a new estimate.

5 Identification against the measure

Green at pass 5 (operator side):

Theorem 5.1 (`B0b, act_symWeighted_eq_smul_act_tilt`). $\text{act } S u = \lambda \cdot \text{act}(\text{tilt}) u$ for $\lambda \neq 0$.

Theorem 5.2 (`B0a, act_iterate_eq_op0f_pow`). $(\text{act } M)^{[n]} u = (\text{opOf } M)^n (\text{toLp } u)$ pointwise.

Theorem 5.3 (`os_reconstruction_uniform_clustering`). *In the window: ONE $m > 0$ such that for EVERY spatial extent L , every connected correlator of the transported transfer operator obeys $|\text{connCorr } T \Omega v n| \leq \|v\|^2 \cdot (e^{-m})^n$. The constant is per-observable; the rate alone is uniform in L .*

The packaging is by citation: `vacuumTransfer_op0f + tiltKernel_symm + clustering_of_gap`, all in-tree. [SLOT: The `gibbsPathSum`-level identification (raw Gibbs two-point sums against the operator powers through B0a/B0b and A3) is the one named remaining step of the charter’s part B; state here exactly what is proved when it lands, and nothing more.]

6 Reproduction

[SLOT: jobs count with predicted-vs-measured table (baseline discrepancy 8366-vs-8480 must be RESOLVED from logs before this row is written); oracle report count; axiom triple; sha256 of every travelling file; fresh-clone protocol (runner `LANE_MODULES` extension).]

7 What is not claimed

No Wightman axioms, no $\text{OS} \rightarrow \text{Wightman}$ analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme’s own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here.